

UNIVERSITY OF THE INCARNATE WORD · FALL 2026 · SECTION 01

# CIS 3370 - Intrusion Detection Systems

Build an intrusion detection pipeline end to end, then reproduce the research that does the same.

**CLASS MEETINGS****Tue & Thu · 3:00–4:15 PM**

Founders Hall G008 · Aug 25 – Dec 3

**INSTRUCTOR****Dr. Gonzalo D Parra**

· (210) 829-3146

**OFFICE HOURS****Mon–Thu · 1:30–3:00 PM**

Founders Hall G009 · scheduled meeting required

**RIGHT NOW**

## This Week

Updates automatically from the Fall 2026 calendar. The matching rows are highlighted in the schedule below.

**Week 3 of 16** Sep 7 – Sep 13**TUE · SEP 8**

NSL-KDD: contents, limits, and feature engineering, with pandas and seaborn (book ch 3.5)

**THU · SEP 10 - TODAY**

Stage 1: data generation, traffic and attack simulation, with tcpdump and nmap

Class meets Tue & Thu 3:00–4:15 PM in Founders Hall G008 · Office hours Mon–Thu 1:30–3:00 PM in Founders Hall G009

## SEMESTER MAP

## Full Schedule

All 16 weeks. The outline is a guideline, changes are announced on Canvas. Filter by what you need.

| WK | DATE       | TOPIC                                                                                     | NOTES                                                                           |
|----|------------|-------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 1  | Tue Aug 25 | Course launch: build an IDS end to end                                                    | Syllabus review · awk warm-up                                                   |
| 1  | Thu Aug 27 | Text processing I: awk as an analyst tool                                                 |                                                                                 |
| 2  | Tue Sep 1  | Text processing II: sed, regex and normalization                                          |                                                                                 |
| 2  | Thu Sep 3  | Log formats: syslog, auth, netflow, JSON                                                  |                                                                                 |
| 3  | Tue Sep 8  | NSL-KDD: contents, limits, and feature engineering, with pandas and seaborn (book ch 3.5) |                                                                                 |
| 3  | Thu Sep 10 | Stage 1: data generation, traffic and attack simulation, with tcpdump and nmap            |                                                                                 |
| 4  | Tue Sep 15 | Stage 2: data collection, with Zeek, Suricata and Elastic Agent                           |                                                                                 |
| 4  | Thu Sep 17 | Stage 3: structuring, from text to fields, with grok, dissect and ECS                     | Capstone paper selection due<br><b>CAPSTONE</b>                                 |
| 5  | Tue Sep 22 | Storage and query: Elasticsearch mappings, search and aggregations                        |                                                                                 |
| 5  | Thu Sep 24 | Group project day                                                                         | Instructor not present · teams work on the capstone<br><b>GROUP PROJECT DAY</b> |
|    |            |                                                                                           |                                                                                 |

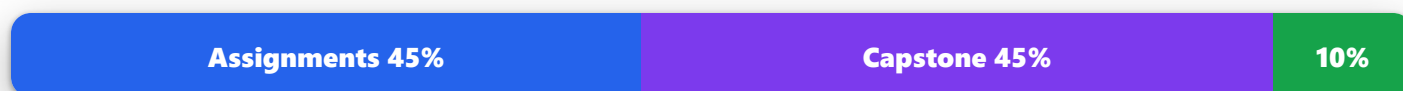
| WK | DATE       | TOPIC                                                                                          | NOTES                                                                           |
|----|------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 6  | Tue Sep 29 | Stage 4: preprocessing, encoding, scaling, imbalance, with scikit-learn (book ch 5.4)          |                                                                                 |
| 6  | Thu Oct 1  | Classical baselines: logistic regression, decision tree and k-NN in scikit-learn (book ch 3.3) |                                                                                 |
| 7  | Tue Oct 6  | Evaluation: the metrics that matter for detection, with scikit-learn (book ch 4.4)             |                                                                                 |
| 7  | Thu Oct 8  | From tabular to time-series: CIC-IDS2017, with pandas windows and a split by time              | Ordering, windows, leakage                                                      |
| 8  | Tue Oct 13 | Sequence models: LSTMs on CIC-IDS2017 (book ch 3.4)                                            |                                                                                 |
| 8  | Thu Oct 15 | <i>No class</i>                                                                                | Fall Break (Oct 15 to 16, university calendar) <b>NO CLASS</b>                  |
| 9  | Tue Oct 20 | Transformers and PEFT on log sequences (book ch 1.3)                                           |                                                                                 |
| 9  | Thu Oct 22 | Capstone progress presentations, day 1                                                         | Feedback session <b>CAPSTONE</b>                                                |
| 10 | Tue Oct 27 | Capstone progress presentations, day 2                                                         | Feedback session <b>CAPSTONE</b>                                                |
| 10 | Thu Oct 29 | Group project day                                                                              | Instructor not present · teams work on the capstone<br><b>GROUP PROJECT DAY</b> |
| 11 | Tue Nov 3  | Reinforcement learning for response selection (book ch 5.3)                                    |                                                                                 |
| 11 | Thu Nov 5  | Stage 6: inference on new data, with a saved model and the agent that runs the pipeline        |                                                                                 |

| WK | DATE       | TOPIC                                                                                  | NOTES                                                                           |
|----|------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| 12 | Tue Nov 10 | Stage 7: alerting, triage and false positives, with Suricata rules and Kibana alerting |                                                                                 |
| 12 | Thu Nov 12 | Stage 8: dashboards and summarizing for humans, with Kibana                            |                                                                                 |
| 13 | Tue Nov 17 | Reading a paper like an engineer, with the capstone proposal form                      | Reproduction method                                                             |
| 13 | Thu Nov 19 | Group project day                                                                      | Instructor not present · teams work on the capstone<br><b>GROUP PROJECT DAY</b> |
| 14 | Tue Nov 24 | Reproduction workshop and capstone Q&A, with the readiness checklist                   |                                                                                 |
| 14 | Thu Nov 26 | <i>No class</i>                                                                        | Thanksgiving holiday (Nov 25 to 27, university closed) <b>NO CLASS</b>          |
| 15 | Tue Dec 1  | Final capstone presentations, day 1                                                    | <b>CAPSTONE</b>                                                                 |
| 15 | Thu Dec 3  | Final capstone presentations, day 2                                                    | Final capstone deliverables due<br><b>CAPSTONE</b>                              |

## HOW YOU'RE GRADED

## Grading Breakdown

Assignments build the pipeline one stage at a time, and those stages become your capstone.



Assignments (one pipeline stage each) - 45%    Capstone reproduction project - 45%

Attendance & Participation - 10%

## Capstone: the 45%, broken down

Reproduce a recent intrusion-detection paper end to end: rebuild its pipeline, reproduce its numbers, or document exactly where it breaks.

|                                                       |     |
|-------------------------------------------------------|-----|
| Paper selection and proposal (approved by instructor) | 2%  |
| Reproduction: working pipeline in your repository     | 15% |
| Final presentation                                    | 10% |
| GitHub repository and project board                   | 8%  |
| GitHub Wiki page: the reproduction report             | 10% |

Missing your scheduled presentation slot costs an automatic **15% off your team's total capstone grade**.

**A failed reproduction is a valid result.** An undocumented one is not. Show what you ran, what you got, and where it diverged.

## Team GitHub requirements

- **Teams of 2 to 3 students, no solo capstones.** Compare schedules before you commit to a team.
- Create a **team GitHub repository**; all advancements and contributions live there.
- Add `gdparra-edu` and `cyberknowledge` as **collaborators**.
- Manage all project activity on the repo's integrated **GitHub Project board**.
- Write the reproduction report as a **GitHub Wiki page**; keep presentation materials in the repo.
- Vote (unanimously) on your **commit approach** and declare it in the proposal: *Individual Commits* or *Delegated Commits*.
- **Group Project Days:** some class sessions are given over entirely to capstone work, marked on the schedule.
- Submit the repo and Project links through the Capstone assignment on **Canvas**.

## WHAT YOU WILL BUILD

# The Pipeline

Every assignment is one stage. By the end of the semester you own all of them.

### Stages, in order

1. **Generate** traffic and logs, including attacks you control.
2. **Collect** them: agents, shipping, storage.
3. **Structure** raw text into fields you can query.
4. **Preprocess**: encode, scale, and deal with class imbalance.
5. **Train** a detector, starting from a baseline you can beat.
6. **Infer** on data the model has never seen.
7. **Alert**, with triage and a false-positive budget.
8. **Dashboard**: put the decision in front of a human.

### Tools and data

- **Text first:** `awk`, `sed`, regular expressions. A log line is text before it is anything else.
- **Datasets:** NSL-KDD first, because it is small and well understood. It is not time-series, so we move to CIC-IDS2017, which has real ordering, before the reproduction work begins.
- **Column reference:** every NSL-KDD column, verified against the files. Ranges recomputed from the data, with the five places the circulating feature table is wrong marked in amber.
- **Models:** classical baselines, LSTMs, transformers, parameter-efficient tuning of language models, and reinforcement learning for response.
- **Everything in Python**, in a repository, reproducible by someone who is not you.

If you cannot rerun it from a clean checkout, it does not count as working.

## THE MAIN EVENT

# Reproduce a Paper

Your capstone is a reproduction study: take a recent intrusion-detection paper and rebuild it end to end.

### How to choose a paper

- Published in the last three years, in a venue with a high h5-index in security and cryptography.
- Check the [Google Scholar venue rankings for computer security and cryptography](#) before you commit.
- It must describe a pipeline you can actually run: data you can obtain, methods described in enough detail to rebuild.
- Get it approved before you start. Two teams may not reproduce the same paper.

### What you hand in

- A repository that runs from a clean checkout.
- A Wiki report: what the paper claimed, what you got, and where the two diverge.
- A presentation to the class.

**A failed reproduction is a valid result.** Report the discrepancy honestly, with evidence. That is how the field works, and it is graded as a success.

**THE FINE PRINT, DIGESTED**

## Policies You'll Actually Ask About

The full versions live in the course outline and syllabus on Canvas - this is the quick reference.

### Late work

- **No late work is accepted.** Every deliverable, assignments and capstone components alike, is due at the stated time. After that the grade for that item is zero.
- This includes the **capstone final deliverables**: the Wiki reproduction report, the repository, the project board and the presentation.
- Documented emergency? Contact Dr. Parra **before** the deadline. Accommodations are arranged in advance, not afterwards.

### Attendance

- Attendance is mandatory and worth **10%**. Up to **3 absences** without penalty, excused or not. The **4th absence** forfeits the entire 10%.
- Arriving **10+ minutes late** counts as tardy; **3 tardies = 1 unexcused absence**.
- Planned university/religious/military absences: written notice at least two weeks ahead. They still count toward the three.

### Academic integrity

Highest standards of academic honesty apply. Cheating and plagiarism are handled per the **UIW Student Handbook**. Using AI tools is expected in this course; passing off work you cannot explain is not.

### This outline is a guideline

The schedule and procedures may change under extenuating circumstances or when it benefits the class. Changes are announced on Canvas.

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**UIW Student Handbook   Student Disability Services   Title IX (Non-Discrimination)**

CIS 3370-01 · Intrusion Detection Systems · Fall 2026 · University of the Incarnate Word · School of Mathematics, Science and Engineering

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